Research in Algebraic Geometry

This is a snippet from the end of *EYNTKA Algebra* [8] that covers areas of research in algebra that a reader may want to explore. Thus, this blog assumes comfort with anything from that book.

It must be pointed out that the topics covered here are *not* a comprehensive list of areas of research, simply large ones. Other areas that are not covered in detail here that have active research include, but are not limited to, the model theory of algebraically and differentially closed fields, probabilistic linear optimization, mathematics of AI (which can be thought of as a continuation of linear algebra and functional analysis), equivalent theory, P vs. NP as an Invariance Theory problem, and so forth.

Algebraic Geometry

The starting point of modern algebraic geometry is the *affine scheme* which generalizes varieties by defining the geometric counter-part to number rings, rings with nilpotents or zero-divisors, rings that are not finitely generated, and overall any commutative ring. These shall be the "model spaces" that are glued together to create *schemes*, where "glued together" is geometrically understood as forming equivalence classes between the points that are being glued and is algebraically understood as adding compatibility relations between functions (essentially, think of the example of projective space where $f \in \mathrm{Hom}(U_1, R), g \in \mathrm{Hom}(U_2, R)$ such that $f(z) = g(1/z)$ on $U_1 \cap U_2$). A lot of the build-up in modern algebra geometry is translating the concepts of differential geometry over $\mathbb{R}$ or $\mathbb{C}$ (ex. smoothness, dimensions, irreducibility, compactness, Hausdorffness, Poincaré duality, monodromy, Riemann-Roch) into the non-analytic setting of schemes. This translation abstracts many of the notions that will require a good deal of build-up, but the reward is a unified language to deal with number theory and varieties which have many of the geometric intuitions one would expect. A lot of similarities shall be shared between differential geometry and scheme theory, but important differences shall also emerge (ex. points containing other points, existence of distinct points that are arbitrarily closed to each other, functions that are not determined by their value at every point).

Scheme are the starting point for the study of many different modern fields, but it can be studied in it of itself. For example to find classification of different types of schemes; this research often involves the use of *moduli spaces* and the theory of *stacks*.

Another very important problem in Algebraic Geometry is the *Hodge Conjecture* which roughly says says that homological information about compact complex manifolds X (which from an algebraic view point, are objects that very closed resemble smooth projective varieties and often are categorically equivalent due to GAGA [11]) can be given by nice subvarieties of X. In particular, without defining any of our terms, it asks which cohomology classes of $H^{k,k}(X)$ can be represented by closed subvarieties Z where X is given the natural Kähler manifold structure. The interested reader wanting to continue the EYTNKA series can read *K-theory* for an introduction to characteristic classes [3], *Algebraic Topology* for the cohomology background [6], *Algebraic Geometry* for the algebraic background [4], and *Differential Geometry* for the Kähler manifold background [9].

Elliptic Curves

The advantage of scheme theory is in full effect when studying curves (Noetherian integral separated 1-dimensional schemes of finite type over k). The study of curves splits to the study of curves of genus g by the Riemann-Roch Theorem (see [10] or [4]). Curves of genus 1 are called *elliptic curves* which by GAGA results ([11]) is equivalent to studying non-singular degree 3 polynomial equations. When in char $\neq 2, 3$ we can algebraically say that the study of elliptic curves is the study of equation $y^2 = x^3 + ax + b$. Finding all rational solutions to such polynomials is still an open problem; the *Birch and Swinnerton-Dyer Conjecture* would, if proven, explain how many solutions there exists, and the person(s) who succeeds to prove it would win $1 million from the Clay Institute of Mathematics[1]. Due to scheme theory there are natural ways of understanding "bad reductions" (think of ramified primes), of defining a covering space using galois theory (like in homotopy theory), and keeping track of when curves intersect non-transversally (with multiplicity higher than 1).

The reader interested can start on *EYNTKA Elliptic Curves* ([2]) or look at *Silvermen's Arithmetic of Elliptic Curves* ([1]).

Number Theory and Arithmetic Geometry

Scheme theory over number fields or number rings is called *Arithmetic Geometry*. A huge amount of progress in modern theory was made in number theory by using tools in arithmetic geometry, for example the proof of Fermat Last Theorem (that there exists no natural number $n \geq 3$ such that $x^n + y^n = y^n$ has an integer solution) was proved through the lens of Arithmetic Geometry and analytic number theory. Much of modern number theory builds directly on the algebraic foundations presented in this book.

A major area of research is the *Langlands program* which aims to generalize Artin's reciprocity (see *Class Field Theory*, [7]) which can be thought of the 1-dimensional case of Langland's reciprocity. The 2-dimensional case strongly involved elliptic curves and the related modular forms and automorphic forms. An introduction to the required background covering these topics includes [2], [4], and [5].

With the advent of complex analysis, many problems in number theory can be studied using certain meromorphic functions. Of central interest is the *Riemann-Zeta function*

$$\zeta(s) = \sum n = 1^{\infty} \frac{1}{n^s} = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}$$

which can be analytically extended to $\mathbb{C} \setminus \{1\}$ and captures an incredible amount of information about the distribution of primes. The *Riemann Hypothesis*, one of the Millennium problems, states that all the [non-trivial][2] roots of $\zeta(s)$ lie on the line $L = \{z \in \mathbb{C} \mid \text{im}(z) = 1/2\}$. By some application of Fourier analysis, this would drastically improve our understand of how many primes $p \leq n$ there exists for any $n \in \mathbb{N}$, and would resolve many open problems. Proving that all [nontrivial] zero's are on L or that there is at least one [nontrivial] zero that is not on L would give the person(s) who proved it $1 million by the Clay Institute.

Geometric Group Theory

Infinite groups are much harder to grasp than finite groups. However there are many infinite groups that naturally show up in geometric settings and have a metric that can be imposed on them. Studying these groups falls under *Geometric group theory* which views infinite groups themselves as geometric objects. By equipping a group with a metric, one can study its large-scale geometry to deduce its algebraic properties. This approach has led to the resolution of major conjectures in topology (most famously, Thurston's Geometrization Conjecture implying the *Poincaré Conjecture*, another Millennium problem).

[1] It is one of the Millennium problems set at the beginning of the 21st century

[2] There are roots that are given by the Gamma function which are considered to be related to the prime at infinity, and is not important for the usual, finite, prime

Derived Algebraic Geometry and Higher Algebra

Schemes are incredibly power, but are not adept with "singular" spaces or when taking "non-regular" quotients; categorically they do not behave well under limit sand colimits. *Derived geometry* is a redevelopment of commutative algebra inspired abstract topology. It replaces rings with derived objects that retain homological information, allowing for a more robust theory of intersections and moduli spaces that works in greater generality. This field often draws on *higher category theory* to create a unified language for geometry.

Motivic and Chromatic Homotopy Theory

Algebraic topology uses algebraic invariants, like homology groups, to classify and distinguish topological spaces. The algebraic tools themselves have become the focus of intense study. *Chromatic homotopy theory* seeks to organize how shapes can be mapped into one another (homotopy) by filtering it through algebraic data coming from the theory of formal groups. *Motivic homotopy theory* builds a direct bridge, or "motive," between algebraic geometry and topology. The ultimate goal is a structural understanding of shape itself, with the calculation of the *stable homotopy groups of spheres* remaining one of the field's major goals.

Anabelian Geometry

Anabelian geometry is asking if we can recover an arithmetic variety X given information about the (étale) fundamental group of X. While Motivic Homotopy Theory seeks to build a dictionary of algebraic invariants to describe a space (using its entire tower of homotopy groups π_n of the sphere spectrum), anabelian geometry exclusively on the first étale homotopy group, $\pi_1^{\text{ét}}$, and posits that for a special class of "hyperbolic" varieties, this single algebraic object already contains all the necessary information.

The central algebraic invariant in this field is the *étale fundamental group*, denoted $\pi_1^{\text{ét}}(X)$. Its connection to the homotopy $\pi^{\mathbf{Top}}$ group comes from how the universal cover $E \to X$ and the intermediate covers has a galois correspondence to the subgroups of $\pi^{\mathbf{Top}}$. An example of a simple étale fundamental group is those for a field k, where $\pi^{\text{ét}}(k) = \text{Gal}_k(k_s)$ where k_s is the maximal separable extension of k. More generally, for a variety X defined over a number field like $\mathbb{Q}$, its étale fundamental group is a profinite group that captures not only the topological shape of X (as seen over the complex numbers) but also the entirety of its arithmetic structure.

The "anabelian" hypothesis is the conjecture that this reconstruction is only possible when the group is sufficiently complex and far from being abelian.

- **Abelian Case (Fails):** For an object like an elliptic curve, whose fundamental group is abelian, this principle fails. Many different elliptic curves can have the same fundamental group, so the group does not uniquely determine the geometry.
- **Anabelian Case (Conjectured to Succeed):** For a "hyperbolic" variety, such as a curve of genus two or more, the étale fundamental group is a non-abelian group. The anabelian hypothesis posits that there is enough information to uniquely determines the original variety.

This makes anabelian geometry a subfield of arithmetic geometry in its goals, but leaning more towards the philosophy of geometric group theory – it seeks to understand geometry by studying the algebraic properties of an associated group. An unsolved problem (as of writing this book) is the *Section Conjecture* which attempts to use the structure of $\pi_1^{\text{ét}}(X)$ to describe the set of rational points on X, offering a new (non-abelian) approach to solving Diophantine equations.

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